

5 (a) (i) $2\pi f = 377$ [M1]
 $f = 60 \text{ Hz}$ [A1]

(ii) $V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{240}{\sqrt{2}}$ [M1]
 $= 170 \text{ V}$ [A1]

(iii) Alternating magnetic field is created in the coil.
Hence there is an alternating magnetic flux linkage through the ring [B1]

By Faraday's law, an e.m.f. will be induced in the ring [B1 – link to emf to flux linkage]

Since the ring is a closed loop, current is induced in the ring. [B1]

By Lenz's law, the current will flow in a direction to oppose the changes in the magnetic flux linkage, hence creating a (average opposing) force which balances the weight. [B1]

(iv) Since the ring is not a closed loop, current is not induced in the ring. [M1]

Hence no opposing force is created, ring does not float. [A1]



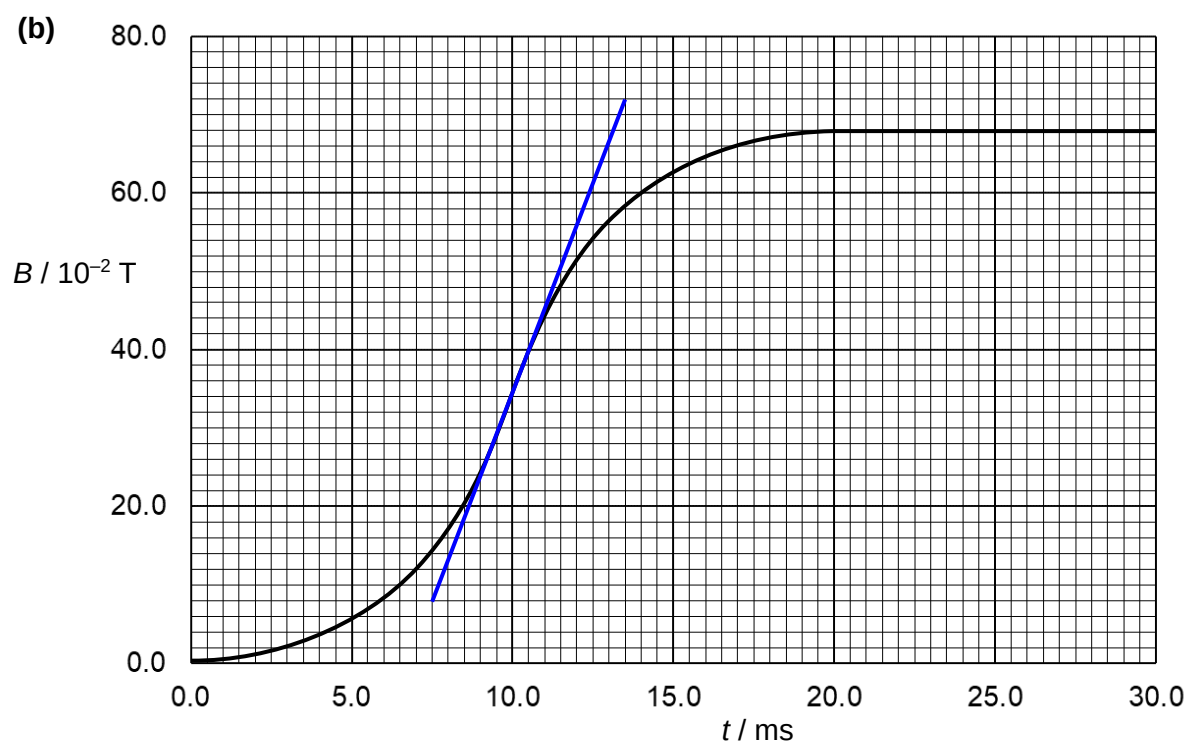


Fig. 5.4

Maximum e.m.f. will be induced when the gradient is steepest, i.e. at 10 ms.

$$\text{Gradient} = \frac{dB}{dt} = \frac{(72.0 - 8.0) \times 10^{-2}}{(13.5 - 7.5) \times 10^{-3}} = 106.67 \quad [\text{M1}]$$

$$|\mathcal{E}_{\text{max}}| = \frac{d(N\phi)}{dt} = NA \frac{dB}{dt} = (80)(4.2 \times 10^{-5})(106.67) = 0.3584 = 0.36 \text{ V} \quad [\text{A1}]$$

Allow variation in value of gradient from 100 to 112 (which relates to a final answer for maximum e.m.f to be from 0.336 to 0.376 V)

–1 mark if gradient triangle is less than one big square (vertically)

6 (a) (i) When the liquid boils, the molecules in the liquid are totally separated